

PHYSICS BOARD EXAM: TOP 50 ESSENTIAL MCQS

Targeted Revision for 2026 | Covers All High-Weightage Modules

MODULE 1: ELECTROSTATICS, POTENTIAL & CAPACITANCE

1. A charge Q is at the center of a cube. Flux through one face is:

- | | |
|---------------------|---------------------|
| (A) Q/ϵ_0 | (B) $Q/6\epsilon_0$ |
| (C) $Q/8\epsilon_0$ | (D) $Q/3\epsilon_0$ |

2. Electric field inside a charged hollow sphere is:

- | | |
|--------------|--------------|
| (A) Zero | (B) E_0 |
| (C) Variable | (D) Infinity |

3. Energy stored in a capacitor _____ when dielectric is inserted (isolated).

- | | |
|---------------|---------------|
| (A) Increases | (B) Decreases |
| (C) Constant | (D) Zero |

4. Which is a polar molecule?

- | | |
|-----------|-----------|
| (A) O_2 | (B) H_2 |
| (C) N_2 | (D) HCl |

5. Work done on an equipotential surface is:

- | | |
|----------|--------------|
| (A) Max | (B) Min |
| (C) Zero | (D) Negative |

6. Plate distance halved, capacitance becomes:

- | | |
|----------|------------|
| (A) Half | (B) Double |
| (C) 4x | (D) 1/4th |

7. Unit of Electric Dipole Moment:

- | | |
|---------|---------|
| (A) C/m | (B) C·m |
| (C) V·m | (D) N/C |

8. Potential at equatorial point of dipole:

- | | |
|---------|------------------|
| (A) Max | (B) Zero |
| (C) Min | (D) constant k |

9. Capacitance of earth is:

- | | |
|-----------------|-----------------|
| (A) 711 μF | (B) 1 F |
| (C) 1 μF | (D) 900 μF |

10. SI Unit of Permittivity:

- | | |
|-----------------------|------------------|
| (A) $C^2N^{-1}m^{-2}$ | (B) Nm^2C^{-2} |
| (C) C/m | (D) Farad |

MODULE 2: CURRENT ELECTRICITY

11. Mobility is drift velocity per unit _____.

- | | |
|-------------|--------------------|
| (A) Current | (B) Electric Field |
| (C) Area | (D) Resistance |

12. Relaxation time in metals _____ with temperature rise.

- | | |
|---------------|---------------|
| (A) Increases | (B) Decreases |
| (C) Constant | (D) Halves |

13. In a non-uniform wire, which stays constant?

- | | |
|---------------------|-------------|
| (A) Current Density | (B) Current |
| (C) Drift Speed | (D) Field |

14. Balance point in Wheatstone bridge means I_g is:

- | | |
|----------|--------------|
| (A) Max | (B) Constant |
| (C) Zero | (D) 1A |

15. SI unit of Conductivity:

- | | |
|----------------|-------------------------|
| (A) Ωm | (B) $\Omega^{-1}m^{-1}$ |
| (C) mho | (D) Siemens |

16. Terminal voltage V is _____ E during charging.

- | | |
|---------|---------|
| (A) $>$ | (B) $<$ |
| (C) $=$ | (D) 0 |

17. Alloys like Nichrome have _____ temperature coefficient.

- | | |
|--------------|----------|
| (A) High | (B) Low |
| (C) Negative | (D) Zero |

18. Drift velocity is of order:

- | | |
|----------------|-------------------|
| (A) 10^8 m/s | (B) 10^{-4} m/s |
| (C) 340 m/s | (D) 1 m/s |

19. Kirchhoff's 1st Law is based on conservation of:

- | | |
|------------|--------------|
| (A) Energy | (B) Charge |
| (C) Mass | (D) Momentum |

20. Resistivity of semiconductor _____ with temperature rise.

- | | |
|---------------|--|
| (A) Increases | (B) Decreases |
| (C) Constant | (D) First $\uparrow$ then $\downarrow$ |

MODULE 3: MAGNETISM, EMI & AC

21. Force between two parallel wires (same dir) is:

- | | |
|---------------|-------------------|
| (A) Repulsive | (B) Attractive |
| (C) Zero | (D) Perpendicular |

22. Dimensions of Self Inductance:

- | | |
|--------------------------|------------------------|
| (A) $[ML^2T^{-2}A^{-2}]$ | (B) $[MLT^{-2}A^{-2}]$ |
| (C) $[ML^2A^{-2}]$ | (D) $[MLT^{-1}]$ |

23. Lenz's Law relates to conservation of:

- | | |
|------------|-----------|
| (A) Charge | (B) Mass |
| (C) Energy | (D) Field |

24. In pure Inductive AC circuit, current _____ Voltage.

- | | |
|----------------------|---------------------|
| (A) Leads by $\pi/2$ | (B) Lags by $\pi/2$ |
| (C) Phase | (D) Lags by π |

25. Lorentz force formula:

- | | |
|--------------------|---------------------|
| (A) $q(v \cdot B)$ | (B) $q(v \times B)$ |
| (C) mv^2/r | (D) qE |

26. Magnetic Field at center of circle:

- | | |
|--------------------|--------------------|
| (A) $\mu_0 I / 2R$ | (B) $\mu_0 I / R$ |
| (C) Zero | (D) $\mu_0 I / 4R$ |

27. Unit of Magnetic Flux:

- | | |
|-----------|------------|
| (A) Tesla | (B) Weber |
| (C) Gauss | (D) Ampere |

28. Soft iron core in MCG makes field:

- | | |
|-------------|--------------|
| (A) Uniform | (B) Radial |
| (C) Zero | (D) Parallel |

29. Transformers work on principle of:

- | | |
|-------------------|-----------------|
| (A) Self Ind. | (B) Mutual Ind. |
| (C) Eddy currents | (D) Hysteresis |

30. Power factor of pure resistive circuit:

- | | |
|---------|--------------|
| (A) 0 | (B) 1 |
| (C) 0.5 | (D) Infinity |

MODULE 4: OPTICS & EM WAVES

31. Fringe width in YDSE immersed in water (4/3):

- | | |
|-----------|-----------|
| (A) 4/3rd | (B) 3/4th |
| (C) Same | (D) Zero |

32. Germ killing EM waves:

- | | |
|--------------|-----------|
| (A) Infrared | (B) UV |
| (C) X-ray | (D) Gamma |

33. Normal adj Telescope Magnification:

- | | |
|---------------|---------------|
| (A) f_o/f_e | (B) f_e/f_o |
| (C) f_o+f_e | (D) D/f_e |

34. Convex mirror image is always:

- | | |
|-------------------|------------------------|
| (A) Real | (B) Virtual/Diminished |
| (C) Real/Enlarged | (D) inverted |

35. Huygens backward wavelet amplitude:

- | | |
|----------|---------------------|
| (A) Max | (B) Zero |
| (C) Half | (D) same as forward |

36. Critical angle is minimum for _____ light.

- | | |
|------------|------------|
| (A) Red | (B) Yellow |
| (C) Violet | (D) Green |

37. Lens maker formula focal length depends on:

- | | |
|----------------------|---------------|
| (A) Refractive Index | (B) Curvature |
| (C) Both | (D) None |

38. Speed of EM waves in vacuum:

- | | |
|-----------|-----------------|
| (A) E/B | (B) B/E |
| (C) EB | (D) $\sqrt{EB}$ |

39. Greenhouse effect is due to:

- | | |
|------------|--------------|
| (A) UV | (B) Infrared |
| (C) X-rays | (D) Radio |

40. Sky appears blue due to:

- | | |
|----------------|----------------|
| (A) Reflection | (B) Scattering |
| (C) Refraction | (D) Dispersion |

MODULE 5: MODERN PHYSICS & SEMICONDUCTORS

41. Ratio of nuclear densities ($A=27$, $A=64$):

- | | |
|-----------|----------|
| (A) 3:4 | (B) 1:1 |
| (C) 27:64 | (D) 9:16 |

42. Bohr radius r proportional to:

- | | |
|-----------|----------------|
| (A) n | (B) n^2 |
| (C) $1/n$ | (D) $\sqrt{n}$ |

43. Forward bias _____ depletion layer.

- (A) Increases
- (C) Constant

- (B) Decreases
- (D) Zeroes

44. Slope of V_0 vs freq plot is:

- (A) h
- (C) e/h

- (B) h/e
- (D) ϕ

45. Isotones have same number of:

- (A) Protons
- (C) Mass

- (B) Neutrons
- (D) Electrons

46. De-broglie wavelength of electron (V):

- (A) $12.27/V$
- (C) $1.227/V$

- (B) $12.27/\sqrt{V}$
- (D) V^2

47. Maximum wavelength transition in Hydrogen:

- (A) $n=2$ to 1
- (C) $n=4$ to 3

- (B) $n=3$ to 2
- (D) ∞ to 1

48. A doped semiconductor is electrically:

- (A) Positive
- (C) Neutral

- (B) Negative
- (D) Magnetic

49. Mass of 1 a.m.u in MeV:

- (A) 931.5
- (C) 0.51

- (B) 1.6×10^{-19}
- (D) 1000

50. PN junction diode as rectifier uses property of:

- (A) High Res.
- (C) Light emission

- (B) One-way cond.
- (D) Doping

OFFICIAL ANSWER KEY

1. B	2. A	3. B	4. D	5. C
6. B	7. B	8. B	9. A	10. A
11. B	12. B	13. B	14. C	15. B
16. A	17. B	18. B	19. B	20. B
21. B	22. A	23. C	24. B	25. B
26. A	27. B	28. B	29. B	30. B
31. B	32. B	33. A	34. B	35. B
36. C	37. C	38. A	39. B	40. B
41. B	42. B	43. B	44. B	45. B
46. B	47. C	48. C	49. A	50. B